

Board to Be Wild

Engineering Design Report

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1 Introduction

In this project I am creating a charcuterie board as a mothers day gift for my mom. We will create the board by milling, cutting, sanding, and gluing together five, two inch pieces of wood. We will create a file with our design in fusion 360 then export it to Aspire to create the tool path for the milling machine. We can make engravings into the wood using either the milling machine or the deep engrave feature of the laser cutter. Once our design is complete we will use resin to fill the engravings that we have made. After the resin has been solidified we will sand down the board to make it smooth. This project will teach us lots of new skills such as how to make tool paths in aspire and how to use the woodworking machines in the Fablab.

2 Initial Design and Planning

My original design idea was to create a board in 3 different pieces that are held together with white resin. Each of the different pieces in the image would have the appearance of different pieces of wood that can be found outside. The biggest problem was the reference image I was looking at was made with three different pieces of wood. If I was to do that design I would have to cut out three different pieces from one board using the milling machine. and if I got three pieces from the same board then the shades of the wood would be the same altering the final product. Even with the impediments I still planed to carry through with my original idea and make the design. I spent multiple class periods trying to refine the design that came from the image I imported because there were lots of edges and turns from the trace that were to sharp for the machine to make. After I fixed all of the turns there

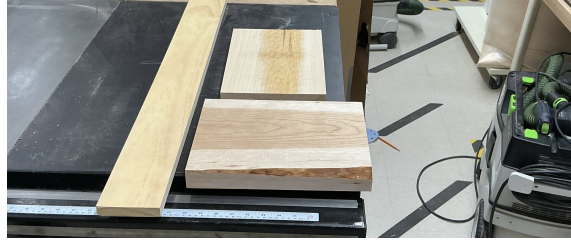


Figure 1: Initial Wood 10.75x11.25x1.13

were lots of incomplete parts in the shape and I could not get it to work. When I had deleted sections trying to remove the sharp turn I had not fully reattached the remaining parts. When I would import the adobe file into Fusion 360 from Adobe Illustrator Fusion would not recognize the different islands as complete shapes. Mrs. Morrow told me that the cutting and resin process would take a long and require me to come in after school, and it would not be finished by mothers day. I so I decided to switch to an easier less time consuming project. My new design was a board with a circular thumb hole cut out in the top left. I would not do any engraving and instead put a thin layer of resin on top. I would layer different shades of blue with a small amount of white to make it appear like a wave crashing onto the beach.

3 Material Preparation

For my preparation I was taken to the outside shed to pick out two different types of wood, cherry and maple, that we would use to make our board. Once we brought the pieces in we used the table saw to cut 5 different 2 inch strips, I got 3 cherry and 2 maple. I glued the different pieces together and then let them sit overnight to dry. Once the board was assembled, we used the planer to make the sides smooth as they were not glued perfectly and not every piece was the same size. We also used the table saw again to make the edges the edges even.

4 Digital Design

When we were first presented with the project, I spent around 20 minutes looking for a board that my mom would love. I settled on a board with three different island like pieces of wood that would be held together with white resin. Imported the file into Adobe illustrator and used the trace tool to get the outline of the pieces of wood. The design had lots of edges and turns that the milling machine would not



Figure 2: My five pieces together

be able to do because the cutting piece was too thick, so I spent lots of time refining the file. In my effort to remove the sharp turns I created lots of pieces that were not completely connected, as a result when I uploaded the design to Fusion 360 the machine was not recognizing the different sections as shapes. I spent lots of time trying to reconnect all of the incomplete shapes but I was fruitless in my efforts. Mrs. Morrow also told me that the creation of the board would take a very long time including time out of class and it would not be done in time for Mothers Day. I decided to change my design so I went back onto the internet and found a board that had resin poured on top in a way that looked like a wave crashing into the sand. My mom loves the beach, so I decided to make that my new design. When creating this design I skipped making a file in Adobe Illustrator because of my past issues and because it would be much quicker and simpler to make the file directly in Fusion 360. I started by creating the board by making a rectangular prism the exact dimensions of the board that I had created. I then used the circle tool to make the area that would be the thumb hole. I wanted to make the board personalized to my mom in some way so I added her initials in the top right corner using the text tool. The text that I used was full of vectors that I spent a long time trying to fix, but once again it was fruitless. I decided to remove the initials and upload my file to aspire. Because I was only doing a profile cut and no engraving the aspire file only took a few moments to make. Mrs. Morrow checked over my file and decided that it was ready to be uploaded to the CNC Machine.

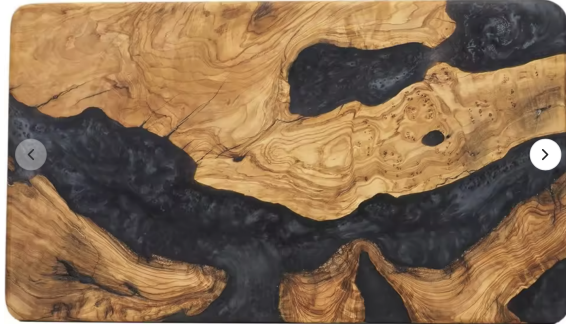


Figure 3: Original idea



Figure 4: Second idea

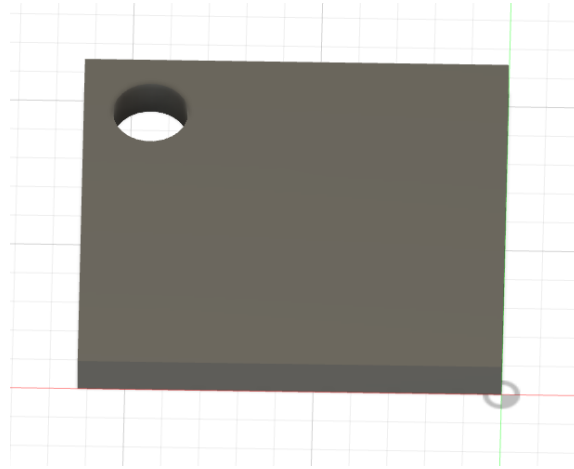


Figure 5: Fusion design

5 CNC and Manufacturing

To use the CNC machine I first downloaded the file off of the google drive and uploaded it to the Shop-bot software. We then put the put double sided tape and secured the board to the machine. once the board was secured and ready to cut we ran a air cut above to make sure that the area we would be cutting was correct. The air cut ensures that the thumb hole I was cutting out doesn't end up in the middle of the board or on the edge. Once the air cut was complete we ran the real cut and it took about 30 seconds. After I had failed to have my moms initials engraved using the Shop-bot machine I decided to try and use the deep engrave tool. I created a file in adobe illustrator and was prepared to do the cut on my board when I realized how long it would take for the deep engrave. With the added time there would be no way I finished my board before mothers day so I scraped the idea.

6 Resin Process

Before I made my resin pour I had to fully sand my whole board. I started with the 60 gain and worked up to the 220 grain. after my board was smooth I started the resin pouring process. My resin pour was going to be especially difficult because there was no specific area or color that I was doing the pour on. I was doing the pour on a flat surface where it would be easy for the resin to go to far and I was using a combination of different blues and white to make the appearance of a wave crashing onto the beach. before pouring any of the resin I first covered my thumb

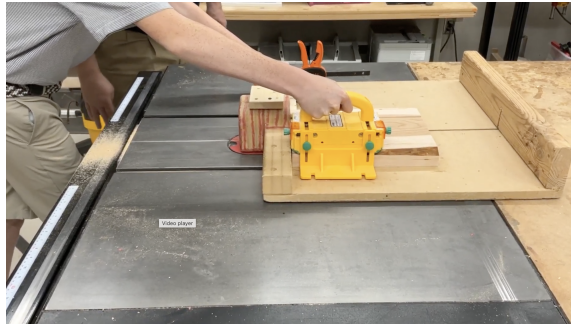


Figure 6: Cutting with table saw video

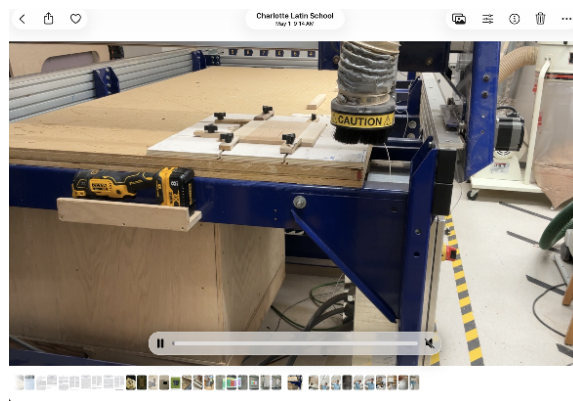


Figure 7: Cutting Video



After

Figure 8: After Cut

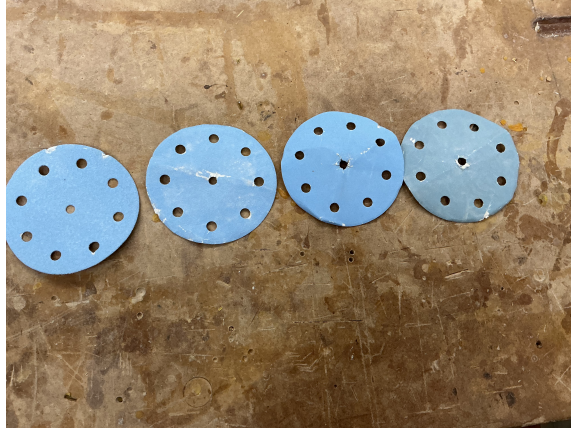


Figure 9: Sand Paper 60, 120, 180, 220



Figure 10: Sanding Board Video



Figure 11: Final Product

hole in tape so none of the resin could get in it. I also elevated the board using plastic cups because if the resin dripped to the cardboard and it was still connected to the board then the board could get stuck to the cardboard. I started pouring the resin by adding the darkest shade of blue in the back corner. I then continuously progressed pouring lighter shades of blue until I got to the middle of the board. I then poured a line of white resin on the edge of all the blue and a few more lines throughout the middle of it. using the heat gun I spread out the white across the blue making the appearance of bubbles in a recently crashed wave. After the resin dried the next day I sanded it again, then soaked the whole board in water to get it more used to liquid and prevent future wilting. After the water had dried I sanded it one last time then applied the oil across the whole board to add a protective coat.

7 Final Product

8 Reflection

My final product came out very nice. The wave worked perfectly and it presented the illusion of a real wave crashing onto the beach. My mother loved the board and it will be used as a charcuterie board. The biggest challenge that I faced was in the processes of making my design. When I was unable to make my original design work it was very frustrating, but I was able to limit the setback and shift to a new design of the wave that I believe was better than the first idea I had. Some of the improvements that I could have made would be to make sure I am always prepared. I messed up because I did not download the file for my cut to the google drive. The

blunder cost me a day and it would have been the difference between me getting the board to my mom on mothers day and a day late. The mistake is a small one, but it will continue to haunt me if I do not change and make sure I am always prepared. In this unit we learned more than anything we have done before. It was my first time doing wood working and just about everything else that we did. We learned how to use just about every tool in the wood working room from the table saw to the sanders. We also learned how to design an aspire file, pour resin, and glue together a board. With the many skills that we learned I think that the most important one will be the experience of using the tools in the wood working room. The other tools are for very specific needs while the tools in the woodworking room will be important for a variety of projects. Overall, I enjoyed working on this unit and had a lot of fun making a personalized board.